Lecture 9 – Lab 9

Algorithm design techniques

Implement Fibonacci number algorithm Fib(n) using:

1- recursive function (brute force)
2- Dynamic programming (top-down approach)
3- Dynamic programming (bottom-up approach)
4- Optimize the Dynamic programming according to space complexity to reach to O(1)

Note for teaching assistant: This is covered at lecture, Let students participate in the idea and implementation (don't discuss everything)

© 2026 Ahmed Elrefaiy

Ex.1: 0/1 Knapsack Problem

– **Example**

- item 1: 7 lbs, $42
- item 2: 3 lbs, $12
- item 3: 4 lbs, $40
- item 4: 5 lbs, $25
- W = 10

How should we pack the knapsack?

1- Solve by Greedy Algorithm, then by dynamic programming
2- if you solve it by brute Force, what is the expected time complexity?
3- apply greedy for the same example as fractional knapsack

n = 4 (# of elements)
W = 5 (max weight)

Elements (weight, Value) is given by:
(2,3), (3,4), (4,5), (5,6)

Weight	Value
2	3
3	4
4	5
5	6

W \ i	0	1	2	3	4
0	0				
1	0				
2	0				
3	0				
4	0				
5	0				

Complete the following table to solve this (0/1) knapsack problem by dynamic programmi